

# ALEXANDER LLEWELLYN SMITH

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## EDUCATION

**Dartmouth College**, Hanover, NH

**EXPECTED GRADUATION: June 2028**

*Bachelor of Engineering*: Mechanical Engineering

*Bachelor of Arts*: Engineering Sciences, Minor: French

Relevant Coursework: Distributed Systems, Thermodynamics, Embedded Systems, Computer-Aided Design & Kinematics, Digital Electronics, Solid Mechanics, Machine Engineering

Activities: Dartmouth Formula Racing, Drummer in student band, Club Snowboard, Chi Gamma Epsilon Fraternity

## RELEVANT EXPERIENCE

**ENGS 33 - Solid Mechanics**, Dartmouth College

**September 2026 – Present**

*Teaching Assistant*

- Hosted office hours for students to go over homework assignments and labs
- Graded homework assignments, labs, and bridge project work
- Supported students in designing, iterating, and building scale wooden bridges for end-of-term project

**ENGS 29 - Computer-Aided Design & Kinematics**, Dartmouth College

**March 2026 – June 2026**

*Teaching Assistant*

- Hosted office hours for students to go over homework assignments and labs
- Graded and managed regrading of homework assignments
- Prepared students for CSWA and CSWP exams

**Commonlands LLC**, San Diego, CA

**June 2025 – August 2025**

*Engineering Intern*

- Designed and 3D printed custom mounts to expedite manufacturing process of camera lenses
- Developed and tested new manufacturing methods to increase efficiency on the assembly line
- Supported setup and ran lens manufacturing machine
- MTF (Modulation Transfer Function) tested lenses to develop detailed product specification sheets for customers
- Strengthened team-building and interpersonal skills within startup company

**Laboratory for Computational Cellular Mechanobiology**,

**May 2023 – July 2024**

Department of Mechanical and Aerospace Engineering, UC San Diego

*Intern*

- Created 3D models to solve differential equations describing cell functions
- Aided in the development of SMART, Spatial Modeling Algorithms for Reactions and Transport
- Created documentation describing detailed research to be used to help onboard new interns
- Worked with Professor Padmini Rangamani and Dr. Emmet Francis

## PROJECTS

**Bipedal Robot**

**June 2025**

*Solo-Project*

- Designed/built a fully functional RC bipedal robot integrating mechanical kinematics with electrical systems in 3D CAD
- Optimized the footpath of a custom Watt six-bar linkage using SolidWorks motion study to create the smallest overall design possible

**Wooden Scale Bridge**

**August 2026**

*Project Lead*

- Designed and tested bridge designs in SAP2000, building final design as a SolidWorks assembly
- Ran Finite Element Analysis and iteratively redesigned bridge based on maximum stress, buckling, and deflection results
- Laser-cut and glued pieces to build bridge and test in Instron machine

## SKILLS & INTERESTS

3D + 2D CAD & Design: SolidWorks CSWA, Autodesk Fusion360, Autodesk Inventor, SAP2000, GD&T, FEA, Motion Study

Mechanical: 3D Printing, Laser Cutting, Laser Engraving

Electrical & Software: Python, Java, C, MATLAB, LaTeX, Embedded Systems, Wiring

Interests: Car spotting, blink-182, snowboarding, Los Angeles Chargers

Languages: English (Native), French (Native), Spanish (Working proficiency)